



One-to-many two-sided matching decision of logistics O2O platform considering the intermediary benefit

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Abstract

In existing logistics online-to-offline (O2O) platform decision activities, psychological behavioral behaviors of logistics demanders and suppliers have become increasingly complex, and the required information also involves a great deal of uncertainty and vagueness. Hence, a two-sided matching (TSM) method considering multiple psychological behavioral behaviors and the intermediary benefit is proposed. First, a one-to-many supply–demand matching problem on online platform with multiple fuzzy preferences is described. To solve this problem, a novel weight-solving algorithm considering subjectivity and objectivity is developed. Matching utility values are determined according to the prospect theory and attribute weights. Based on desired matching numbers between logistics supply and demand agents and matching utility values, a unilateral expected matching model is constructed. Then, the expected matching matrix is obtained. By using regret theory, agent satisfaction is calculated according to the expected matching matrix. A novel formula considering risk preference coefficients is developed to calculate the intermediary benefit. Moreover, a one-to-many TSM model is established to maximize agent satisfactions of two-sided agents and the intermediary benefit. To obtain a reasonable logistics service matching scheme, a novel max–min algorithm considering fairness is developed. Finally, the rationality, effectiveness and practicality of the proposed method are verified through a one-to-many matching example on a logistics O2O platform.

Keywords Two-sided matching · Logistics O2O platform · Multiple psychological behavioral factors · Intermediary benefit · Multiobjective programming

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1 Introduction

E-commerce is one of the fast-growing industries in recent years. The surging logistics demand due to the rapid development of e-commerce market elevates the intensity and complexity of logistics activities, especially for the parcel/freight delivery within urban areas, and brings enterprises and cities new challenges [1]. In particular, the evolving characteristics of e-commerce reshape the third-party logistics industry, in which more third-party-forwarding logistics services emerge and have received increasing popularity in recent years [2].

Logistics is a crucial element in metropolitan areas' economic competitiveness and sustainability, and freight transportation takes the role of facilitator by connecting supply and demand in a dynamic market [3]. To fulfill increasing demands of economic development, China has constructed extensive road freight infrastructure to enhance its freight ability and capacity [4]. Therefore, this paper is based on O2O logistics information platform, with the aim of exploring the optimal matching decision between supply and demand. To achieve this goal, two-sided agents need to have convenient and real-time information exchange. However, in actual freight situations, there are often issues such as high transaction costs, information asymmetry, or communication difficulties. Kanda et al. [5] mentioned that intermediaries can connect participants with their relevant activities, skills, and resources in situations where high transaction costs, information asymmetry, or communication issues make direct interaction difficult. Virtanen et al. [6] provided that intermediaries often possess knowledge and access to local business relationships, and suggested that using intermediaries can reduce various costs such as setting up distribution networks. The O2O logistics information platform studied in this paper can be regarded as an intermediary platform, providing matching services for both supply and demand sides through information search technology, reducing tedious intermediate links and service costs.

With the surge in demand for demand-based O2O services, Chinese instant logistics industry has experienced explosive growth in recent years, leading to a greater diversity in online shopping options [7]. Among them, the freight O2O platform is based on online taxi-hailing platforms, and it is flourishing in China. There exist more than 200 freight O2O platforms in operation [4]. O2O freight platforms are very likely to replace the traditional intermediary logistics companies by directly linking the truck owners with consumers [4]. There are already successful O2O platforms in China, such as Didi in the taxi and car sharing business, and Huolala in the light-duty truck business [8].

Currently, O2O logistics information platforms are rapidly developing, expanding their service scope and content in more refined directions. Their services are gradually evolving from basic vehicle-freight matching to various forms, such as logistics dedicated lines and international logistics. However, there are still several remaining challenges that need to be addressed. First, when conducting transactions through O2O logistics information platforms, both the supply and demand sides still need to manually screen and find potential matches, which requires significant time and material costs. Second, the matching mechanism of O2O logistics information platforms relies mainly on the provided vehicle and goods information, without considering the remaining information of the matching agents, resulting in a certain level of mistrust between the supply and demand sides. Third, the current simple vehicle-freight matching method based on the provided requirements and expectations can maximize satisfaction only for individual agents and lacks a holistic approach to maximize satisfaction for both the supply and demand sides. Finally, psychological behavioral factors such as risk aversion may influence the matching process between the supply and demand sides, which is not considered in the existing O2O logistics information platform matching mechanisms.

Two-sided matching (TSM) decision refers to the process of seeking optimal matching agents for two mutually exclusive sets of agents based on the evaluation information provided by two-sided agents and the decision-maker. In real-world two-sided matching decision scenarios, two-sided agents need to provide their evaluation information, and based on this information, different types of third-part intermediaries such as individuals, companies, and governments can find suitable matching agents for the two participating agents. The ultimate goal is to form a satisfactory and stable matching solution for two-sided agents. By publishing vehicle and goods information on O2O logistics information platforms, which operate in a voluntary market environment with non-crossing cooperation between the supply and demand sides, the mutual choice between vehicle owners and cargo owners can be viewed as a two-sided matching problem. Furthermore, as an effective type of third-part intermediary, O2O logistics information platforms aim to maximize the overall benefits of the supply and demand sides by comprehensively considering the actual needs and levels of vehicle owners and cargo owners, providing reasonable and satisfactory matching solutions, and promoting the rational allocation of relevant logistics resources in society while improving logistics operation efficiency in the production flow process. Therefore, researching the vehicle-freight two-sided matching problem in O2O logistics information platforms is of significant practical significance in terms of improving platform returns and increasing satisfactions of two-sided agents.

In summary, this paper proposes a decision method for solving the one-to-many supply–demand matching problem on logistics O2O platforms, taking into account a mixed fuzzy environment involving multiple psychological behavioral factors and the intermediary benefit. The research plan is as follows: On the basis of using a cloud model to process linguistic term set information, a method combining the best–worst method (BWM) with the maximum deviation method is proposed to determine the weights of attributes. The TDOIM (a Portuguese acronym denoting Interactive and Multicriteria Decision-Making) is employed to calculate the matching utility value and obtain the expected matching matrix. Considering the regret aversion psychology of matching participants in the second round of matching, the satisfaction degrees of the participants are calculated based on regret theory. Considering the intermediary benefit objective and one-to-many matching constraints, a multi-objective programming model is constructed, and a novel max–min algorithm that considers fairness is designed to solve the model and obtain the matching solution for the logistics O2O platform.

The remainder of this paper is structured as follows: In Sect. 2, related literature on the cloud model, the hesitant fuzzy linguistic term set (HFLTS), and TSM is discussed. The one-to-many matching problem considering the intermediary benefit is described in Sect. 3. Section 4 provides a one-to-many TSM behavioral decision considering the intermediary benefit and various psychological behavioral factors. In Sect. 5, a case study is employed to verify the effectiveness of the proposed method. Section 6 summarizes this paper.

2 Literature review

2.1 Two-sided matching

TSM decision refers to the process of finding the optimal matching scheme between two disjoint sets of agents based on their preferences, through a well-defined matching mechanism. The origin of TSM theory can be traced back to the research by Gale and Shapley [9] on the marriage-matching problem. Building upon this foundation, scholars have applied TSM

theory to solve other decision problems in real-life scenarios. For various TSM problems, corresponding decision methods have been developed, such as the job-market matching problem [10], product manufacturing matching problem [11], and so on. In the context of O2O logistics platform, the TSM problem requires both logistics service providers and customers to provide preference evaluation information based on their specific needs. However, in the process of transactions between logistics service providers and customers through the O2O platform, due to the complexity of the platform transaction environment and the fuzziness of decision-makers' thinking, single or mixed fuzzy information representation tools such as hesitant fuzzy sets, intuitionistic fuzzy sets, interval intuitionistic fuzzy sets, and linguistic term sets are more suitable for describing the decision evaluation information provided by the logistics service providers and customers. Therefore, some scholars have conducted research on the TSM decision problem based on single or mixed fuzzy preferences. For example, Tong and Zhu studied the TSM problem in a fuzzy environment for cloud manufacturing and proposed a TSM decision method in a linguistic preference environment [12]. Xiang et al. addressed the TSM problem in high-end equipment manufacturing and proposed a new TSM decision method based on hesitant fuzzy sets [13]. Jia et al. developed a novel TSM method for solving the TSM problem with probabilistic linguistic term sets [14]. Chen et al. presented a TSM decision method for personnel rescue problems that considering multiple fuzzy preference information, including linguistic variables and intervals [15]. Wu et al. proposed a TSM decision method incorporating various fuzzy preference information, such as linguistic term sets and intervals [16].

Meanwhile, as a third-party intermediary, the platform's involvement has a significant impact on the formation of TSM solutions. Therefore, scholars have also conducted in-depth research and discussion on the decision problem of the TSM involving third-party intermediaries. For example, Aoyagi and Yoo studied the impact of e-commerce platforms on the TSM decision [17]. Pu and Yuan proposed a TSM decision method considering the third-party intermediary's benefit for knowledge innovation management problems [18]. Liang et al. presented a TSM decision method considering the third-party intermediary's benefit for personnel-machine matching problems under interval intuitionistic fuzzy preferences [19]. Furthermore, psychological behavioral factors of the transaction agents also play a important role in the matching process of the O2O logistics platform. Therefore, some scholars have explored the impact of psychological behavioral factors of two-sided agents on the matching solutions. For instance, Zhao et al. tackled the ridesharing system problem and constructed a TSM model based on the regret theory [20]. Liang et al. proposed a one-to-one TSM decision method based on prospect theory and a collaborative filtering algorithm for the TSM problem in an incomplete preference environment [21]. Zhang et al. introduced a new TSM decision method for an incomplete fuzzy preference environment using regret theory [22]. Zhang et al. proposed a TSM decision method with multigranular hesitant fuzzy linguistic term sets [23]. It can be seen that the existing literature has made significant research achievements in the TSM decision problems considering fuzzy preferences, third-party intermediaries, and psychological behavioral factors. However, there is a scarce research on the TSM problems in mixed fuzzy environments involving multiple psychological behavioral factors and the intermediary benefit.

2.2 Logistics O2O platform

The concept of a "logistics platform" was first proposed by Aldin and Stahre [24], who conducted case studies and found that integration of e-commerce and logistics led to the

emergence on logistics platforms in certain industries. Based on the theory of two-sided markets, Lee pointed out that monopoly-type logistics platform companies have different pricing strategies [25]. Deng et al. addressed the problem of unbalanced competition in logistics information platform resources and proposed a two-stage optimization method for resource allocation [26]. Qiu et al. found through research on cross-border supply chains that the best strategy for government subsidy withdrawal utilizes market-oriented logistics platforms [27]. Zhen et al. studied the vehicle scheduling problem on a hybrid logistics platform and proposed a new algorithm for solving the integer programming model constructed [28]. Zhang and Ma's research results indicated that the level of logistics service platforms has a certain impact on consumers' purchasing behavior [29]. Based on an analysis of data from a Belgian transport platform, Rai et al. found that the use of logistics platforms leads to higher external transportation costs [30]. Lu et al. studied heterogeneous delivery tasks on a hybrid logistics platform and constructed a new mixed integer programming model [31]. Qin et al. presented a model for sharing e-commerce logistics platforms [32]. Tufano et al. developed a data-driven logistics information platform and applied it to the operations of the port of Rotterdam [33].

With the maturity and implementation of mobile internet, O2O as a business model that effectively combines online traffic and offline services has received significant attention from many businesses and has been applied in various industries. Current research on O2O platforms mainly focuses on two aspects: operational strategies under the O2O model and optimizing user experiences on O2O platforms. Wang et al. conducted research on consumer evaluation behavior patterns on O2O platforms using social network analysis methods and found that online product prices, product quality, and business reputation significantly influence consumer decision on O2O platforms [34]. Ma et al. studied the O2O tourism supply chain under the influence of big data and found that altruistic preferences can effectively promote the sustainable development of low-carbon tourism supply chains [35]. Liang et al. constructed an evaluation index system for customer satisfaction on O2O food delivery platforms and proposed an evaluation method based on probabilistic linguistic term sets [36]. Wu et al. examined the low-carbon decision problem in online shopping under the O2O model, considering the characteristics of the O2O model and consumers' low-carbon awareness [37]. Fan et al. investigated the causes of dangerous driving behavior among delivery drivers in O2O food delivery platforms and found that government propaganda and inspections have a certain influence on drivers' dangerous driving behavior [38].

This paper draws inspiration and insights from previous research on the TSM decisions and logistics O2O platform. First, research on logistics O2O platforms primarily focuses on the construction, operation, and pricing of logistics O2O platforms, while there is relatively less research on the matching mechanism of logistics O2O platforms. Second, most of the existing research has provided two-sided matching methods that consider the single psychological behavioral factor, while there is little consideration for situations where decision-makers exhibit multiple psychological behavioral factors. The comparative analysis among the related literature and this paper is displayed in Table 1.

2.3 Motivations and contributions of this paper

After reviewing the above research on TSMs and logistics O2O platforms, there are still the following shortcomings: (1) There is a lack of research exploring the TSM of logistics O2O platforms with multiple fuzzy preference information. (2) The method for solving attribute weights under multiple fuzzy preference environments is still not perfect. (3) The methods

Table 1 Comparative analysis of related literature with the proposed methodology

Method	Fuzzy preference information	Attribute weight	Psychological behavioral behaviors	Intermediary	Model	Model solving algorithm
Literature [12]	Hesitant fuzzy set	Unknown	Learning and synergy effect	No	Yes	Improved hybrid algorithm
Literature [13]	Hesitant fuzzy set	Unknown	No	No	Yes	Enhanced gray wolf algorithm
Literature [14]	Hesitant fuzzy set	Unknown	No	No	Yes	Novel Pareto refining method
Literature [21]	Incomplete preference ordinal	Unknown	Prospect theory	No	No	No
Literature [18]	Numerical value, interval number, Unordered enumeration	Unknown	Prospect theory	Yes	Yes	Linear weighted method
Literature [15]	Triangular fuzzy number, linguistic term set, preference order	Unknown	Prospect theory	Yes	Yes	Single objective optimization
Literature [16]	Intuitionistic fuzzy set	Unknown	Prospect theory	Yes	Yes	Two stage method
The methodology proposed in this paper	HFLTS, real number, triangular fuzzy number and intuitionistic fuzzy number (IFN)	Unknown	Prospect theory, regret theory	Yes	Yes	Max–min optimal algorithm considering fairness

for calculating the intermediary benefit in TSM studies are inadequate. (4) Existing research on TSM models has relatively little consideration for fairness among agents. (5) Finally, the multiple psychological behavioral factors have yet to be developed.

Therefore, motivations of this paper are summarized as follows:

- (1) There is significant value in conducting research on the TSM decisions of logistics O2O platforms considering multiple fuzzy preference information.
- (2) A reasonable method for calculating attribute weights holds great importance in determining the TSM scheme.

- (3) Given the crucial role played by intermediaries in the logistic O2O platform TSM process, it is necessary to incorporate the intermediary benefit into TSM decision.
- (4) It is necessary to construct a TSM model considering multiple psychological behavioral factors.
- (5) Designing a novel algorithm considering fairness is practical for the constructed TSM model.

Among them, main contributions of this paper are demonstrated as follows:

- (1) A TSM method on a logistics O2O platform that accounts for multiple psychological behavioral factors is presented.
- (2) A novel algorithm considering subjectivity and objectivity for determining attribute weights is designed.
- (3) A method for calculating the intermediary benefit considering risk preference and matching willingness is presented.
- (4) A one-to-many TSM model considering multiple psychological behavioral factors is developed.
- (5) A max–min optimal algorithm considering the fairness principle is designed.

3 Basic knowledge

3.1 Cloud model

The cloud model is an innovative approach to qualitative language processing that is designed to capture the inherent vagueness and uncertainty of human thought in the real world, effectively achieving the natural transformation between qualitative language and quantitative values.

Definition 1 [39] Given a qualitative concept Q in domain G , if any element $x \in G$ satisfies $x \sim N(Ex, En'^2)$ and $En' \sim N(En, He^2)$, the membership degree $r(x)$ of x belonging to G satisfies the following conditions:

$$r(x) = e^{-\frac{(x-Ex)^2}{2En^2}} \quad (1)$$

The mapping distribution from domain G to interval $[0, 1]$ $[0, 1]$ is referred to as a Gaussian cloud, where x is the cloud drop. Cloud models are typically described using three numerical features Ex , En , and He to represent their randomness and fuzziness. Here, Ex represents the mathematical expectation of cloud droplets in the domain and is also the most representative point value in the domain; En represents the acceptable dispersion degree of cloud droplets in the domain and serves as a measure of uncertainty in the cloud model; and He represents the thickness of the cloud model and the uncertainty degree of entropy. For convenience, we denote $z = (Ex, En, He)$. If the mathematical expectation in the cloud model z is an interval number $[\underline{Ex}, \overline{Ex}]$, then $z = ([\underline{Ex}, \overline{Ex}], En, He)$ is called a Gaussian trapezoidal cloud.

Definition 2 [40] Given two Gaussian trapezoidal clouds $z_1 = ([\underline{Ex}_1, \overline{Ex}_1], En_1, He_1)$ and $z_2 = ([\underline{Ex}_2, \overline{Ex}_2], En_2, He_2)$, the distance between the two clouds can be computed by

$$d(z_1, z_2) = \sqrt{\frac{1}{4} \left[\frac{(\underline{Ex}_1 - \underline{Ex}_2)^2 + (\overline{Ex}_1 - \overline{Ex}_2)^2}{(En_1 - En_2)^2 + (He_1 - He_2)^2} \right]} \quad (2)$$

Definition 3 [40] Given two Gaussian trapezoidal clouds $z_1 = ([\underline{Ex}_1, \overline{Ex}_1], En_1, He_1)$, $z_2 = ([\underline{Ex}_2, \overline{Ex}_2], En_2, He_2)$; then $z_1 = ([\underline{Ex}_1, \overline{Ex}_1], En_1, He_1)$ and $z_2 = ([\underline{Ex}_2, \overline{Ex}_2], En_2, He_2)$ can be transformed into interval numbers through 3 En principle. Furthermore, the comparative rules between $z_1 = ([\underline{Ex}_1, \overline{Ex}_1], En_1, He_1)$ and $z_2 = ([\underline{Ex}_2, \overline{Ex}_2], En_2, He_2)$ are represented as follows:

- (1) if $L_{\tau v} > 0$, then $z_1 > z_2$;
- (2) if $L_{\tau v} = 0$, $En_1 < En_2$, then $z_1 > z_2$;
- (3) if $L_{\tau v} = 0$, $En_1 = En_2$ and $He_1 < He_2$, then $z_1 > z_2$;
- (4) if $L_{\tau v} = 0$, $En_1 = En_2$ and $He_1 = He_2$, then $z_1 = z_2$,

where $L_{n_1 n_2} = 2(\overline{n_1} - \underline{n_2}) - (\overline{n_1} - \underline{n_1} + \overline{n_2} - \underline{n_2})$, $\overline{n_1} = \overline{Ex_1} + 3En_1$, $\underline{n_1} = \underline{Ex_1} - 3En_1$, $\overline{n_2} = \overline{Ex_2} + 3En_2$, $\underline{n_2} = \underline{Ex_2} - 3En_2$.

3.2 Hesitant fuzzy linguistic term set

Definition 4 [41] Let $O = \{o_i | i = -\iota, \dots, 0, 1, 2, \dots, \iota\}$ be a linguistic term set (LTS) with $2\iota + 1$ granularity, where o_i represents the i th linguistic variable. Each linguistic term o_i needs to satisfy the following conditions: 1) Orderliness: if $i < k$, then $o_i < o_k$; 2) negation operator: $\text{neg}(o_k) = o_{-k}$.

Definition 5 [41] Given a LTS $O = \{o_i | i = -\iota, \dots, 0, 1, 2, \dots, \iota\}$, the contiguous subset of LTS O is referred to as a HFLTS, denoted as $H_o = \{o_k | \kappa = 1, 2, \dots, \#H_o\}$. Here, $\#H_o$ refers to the number of linguistic term sets in HFLTS H_o , and o_k refers to the κ th smallest linguistic term.

3.3 Gaussian trapezoidal cloud

Definition 6 [42] Given a LTS $O = \{o_i | i = -\iota, \dots, 0, 1, 2, \dots, \iota\}$, the corresponding linguistic scale function can be expressed as follows:

$$\theta_i = \begin{cases} \frac{u^i - u^{-i}}{2u^i - 2}, & -\iota \leq i < 0 \\ \frac{u^i + u^{-i} - 2}{2u^i - 2}, & 0 \leq i \leq \iota \end{cases} \quad (3)$$

where the values of θ_i are within the range $[0, 1]$, and the values of u can be determined through experimental and subjective methods. Using the experimental method, the range of u is determined to be $[1.36, 1.4]$; using the subjective method, the range of u is determined to be $u \approx 1.37$ [30]. Considering the objectivity of the decision process, in this paper, an experimental method is adopted to determine the range of $[1.36, 1.4]$.

Definition 7 [42] Given a LTS $o_i \in O$, a cloud model $z_i = (Ex_i, En_i, He_i)$ and a domain $[X_{\min}, X_{\max}]$, a linguistic term set $o_i \in O$ can be transformed into a cloud model $z_i = (Ex_i, En_i, He_i)$ through the following process:

Step 1: Using Eq. (3), calculate the linguistic scale θ_i corresponding to the linguistic term o_i .

Step 2: Using Eq. (4), calculate the mathematical expectation Ex_i :

$$Ex_i = X_{\min} + \theta_i (X_{\max} - X_{\min}) \quad (4)$$

Step 3: Using Eqs. (5, 6), calculate the entropy En_i .

$$En'_i = \begin{cases} \frac{(1-\theta_i)(X_{\max}-X_{\min})}{3}, & -\iota \leq i \leq 0 \\ \frac{\theta_i(X_{\max}-X_{\min})}{3}, & 0 < i \leq \iota \end{cases} \quad (5)$$

$$En_i = \begin{cases} \frac{(\theta_{|i|-1} + \theta_{|i|} + \theta_{|i|+1})(X_{\max}-X_{\min})}{9}, & 0 < |i| \leq \iota - 1 \\ \frac{(\theta_{|i|-1} + \theta_{|i|})(X_{\max}-X_{\min})}{6}, & |i| = \iota \\ \frac{(\theta_i + 2\theta_{i+1})(X_{\max}-X_{\min})}{9}, & i = 0 \end{cases} \quad (6)$$

Step 4: Using Eq. (7), calculate the hyper entropy He_i :

$$He_i = \frac{En^+ - En_i}{3} \quad (7)$$

where $En^+ = \max_i \{En'_i\}$.

Definition 8 [42] Given a linguistic term set $O = \{o_i | i = -\iota, \dots, 0, 1, 2, \dots, \iota\}$ and the cloud $z_i = (Ex_i, En_i, He_i)$ generated from the linguistic term s_i , the HFLTS $H_o = \{o_\kappa | \kappa = 1, 2, \dots, \#H_o\}$ based on the linguistic term set O can be transformed into a Gaussian trapezoidal cloud $z_{H_s} = ([Ex_{H_s}, \overline{Ex}_{H_s}], En_{H_s}, He_{H_s})$ through the following steps:

Step 1: Using Eqs. (8, 9), calculate the interval of the expected mean $[Ex_{H_s}, \overline{Ex}_{H_s}]$:

$$Ex_{H_s} = \min_{\kappa=1,2,\dots,\#H_s} \{Ex_\kappa\} \quad (8)$$

$$\overline{Ex}_{H_s} = \max_{\kappa=1,2,\dots,\#H_s} \{Ex_\kappa\} \quad (9)$$

Step 2: Using Eq. (10), calculate the entropy En_{H_s} :

$$En_{H_s} = \sqrt{\sum_{\kappa=1}^{\#H_o} \frac{En_\kappa^2}{\#H_o}} \quad (10)$$

Step 3: Using Eq. (11), calculate the hyper entropy He_{H_s} :

$$He_{H_s} = \sqrt{\sum_{\kappa=1}^{\#H_o} \frac{He_\kappa^2}{\#H_o}} \quad (11)$$

3.4 Two-sided matching

Suppose that $R = \{R_1, R_2, \dots, R_a, \dots, R_p\}$ and $T = \{T_1, T_2, \dots, T_b, \dots, T_q\}$ are two agents sets of logistic suppliers and logistics demanders, respectively, where R_a the a th logistic supplier and T_b represents the b th logistics demander.

Definition 9 [43, 44] Let $\delta : R \cup T \rightarrow R \cup T$ be a mapping; if $\forall R_a \in R, \forall T_b \in T$, η satisfies five constraints: 1) $\delta(R_a) \subseteq R_a \cup T$; 2) $\delta(T_b) \in R \cup \{T_b\}$; 3) if $\delta(R_a) = \{T_{b1}, T_{b1}, \dots, T_{bj}\}$, then $\delta(T_{b1}) = R_a, \dots, \delta(T_{bj}) = R_a$; 4) if $\delta(T_b) = R_a$, then $T_b \in \delta(R_a)$; and 5) $\delta(R_a) \cap \delta(R_l) = \emptyset$, $\forall l \in A, a \neq l$; then, δ is called a TSM. Here, $\delta(R_a) = T_b$ (matching pair (R_a, T_b)) indicates that R_a and T_b are matched successfully in TSM δ ; $\delta(T_b) = T_b$ indicates that T_b is not matched in TSM δ .

4 Multi-index two-sided matching problem considering multiple hesitant fuzzy elements

This paper studies the one-to-many supply–demand matching problem on logistics O2O platforms. The detailed problem description is given as follows: Let $R = \{R_1, R_2, \dots, R_a, \dots, R_p\}$ be the set of logistics suppliers and $T = \{T_1, T_2, \dots, T_b, \dots, T_q\}$ be the set of logistics demanders. The evaluation attribute set that logistics suppliers use to evaluate logistics demanders is denoted as $M = \{M_1, M_2, \dots, M_s, \dots, M_e\}$, where M_s represents the s th evaluation attribute; its corresponding weight set is $w = \{w_1, w_2, \dots, w_s, \dots, w_e\}$. Similarly, the evaluation attribute set that logistics demanders use to evaluate logistics suppliers is denoted as $N = \{N_1, N_2, \dots, N_f, \dots, N_u\}$, where N_f represents the f th evaluation attribute; its corresponding weight set is $\omega = \{\omega_1, \omega_2, \dots, \omega_f, \dots, \omega_u\}$.

Furthermore, the evaluation attribute sets $M = \{M_1, M_2, \dots, M_s, \dots, M_e\}$ and $N = \{N_1, N_2, \dots, N_f, \dots, N_u\}$ provided by logistics suppliers and logistics demanders are further divided as follows: Let the evaluation attribute set of logistics suppliers to logistics demanders be $\Lambda = \Lambda^1 \cup \Lambda^2 \cup \Lambda^3 \cup \Lambda^4$, and the evaluation attribute set of logistics demanders to logistics suppliers be $\Upsilon = \Upsilon^1 \cup \Upsilon^2 \cup \Upsilon^3 \cup \Upsilon^4$; here, Λ^g and Υ^g represent evaluation information in four forms, including HFLTSSs, real numbers, triangular fuzzy numbers, and intuitionistic fuzzy numbers (IFNs). Qualitative attributes are generally represented using HFLTSSs and IFNs, while quantitative attributes are typically represented using real numbers and triangular fuzzy numbers. Let Λ^c and Λ^d be benefit and cost attributes, respectively, in Λ ; Υ^c and Υ^d represent benefit and cost attributes, respectively, in Υ .

For index M_s , let the expected evaluation value provided by R_a be χ_a^s , where χ_a^s can be represented as Eq. (12). Let the expected evaluation value given by T_b under M_s be γ_b^s , where γ_b^s are expressed as Eq. (13). For index N_f , let the actual evaluation value provided by R_a be χ_a^f , where χ_a^f can be represented as Eq. (14). Let the expected evaluation value given by T_b under index N_f be γ_b^f , which is expressed as Eq. (15). The problem studied in this paper is as follows. When considering the intermediary benefit, how can a many-to-one TSM model be built and solved based on multiattribute fuzzy information provided by both sides to obtain a reasonable logistics O2O platform TSM scheme.

$$\chi_a^s = \begin{cases} H_a^s = \{o_a^{sk} | k = 1, 2, \dots, \#H_a^s\} & \text{if } M_s \in \Lambda^1 \\ y_a^s = \rho_a^s & \text{if } M_s \in \Lambda^2 \\ k_a^s = (\underline{c}_a^s, c_a^s, \bar{c}_a^s) & \text{if } M_s \in \Lambda^3 \\ \vartheta_a^s = \langle \mu_a^s, \nu_a^s \rangle & \text{if } M_s \in \Lambda^4 \end{cases} \quad (12)$$

$$\gamma_b^s = \begin{cases} H_b^s = \{o_b^{s\kappa} | \kappa = 1, 2, \dots, \#H_b^s\} & \text{if } M_s \in \Lambda^1 \\ y_b^s = \rho_b^s & \text{if } M_s \in \Lambda^2 \\ k_b^s = \left(\underline{c}_b^s, c_b^s, \bar{c}_b^s \right) & \text{if } M_s \in \Lambda^3 \\ \vartheta_b^s = \langle \mu_b^s, \nu_b^s \rangle & \text{if } M_s \in \Lambda^4 \end{cases} \quad (13)$$

$$\chi_a^f = \begin{cases} J_a^f = \{o_a^{f\kappa} | \kappa = 1, 2, \dots, \#H_s\} & \text{if } N_f \in \Upsilon^1 \\ y_a^f = \rho_a^f & \text{if } N_f \in \Upsilon^2 \\ k_a^f = \left(\underline{c}_a^f, c_a^f, \bar{c}_a^f \right) & \text{if } N_f \in \Upsilon^3 \\ \vartheta_a^f = \langle \mu_a^f, \nu_a^f \rangle & \text{if } N_f \in \Upsilon^4 \end{cases} \quad (14)$$

$$\gamma_b^f = \begin{cases} J_b^f = \{o_b^{f\kappa} | \kappa = 1, 2, \dots, \#H_b^f\} & \text{if } N_f \in \Upsilon^1 \\ y_b^f = \rho_b^f & \text{if } N_f \in \Upsilon^2 \\ k_b^f = \left(\underline{c}_b^f, c_b^f, \bar{c}_b^f \right) & \text{if } N_f \in \Upsilon^3 \\ \vartheta_b^f = \langle \mu_b^f, \nu_b^f \rangle & \text{if } N_f \in \Upsilon^4 \end{cases} \quad (15)$$

5 One-to-many two-sided matching decision considering the intermediary benefit

5.1 Normalization of fuzzy preference information

For ease of calculation, it is necessary to normalize the actual evaluation values and expected evaluation values provided by the logistics suppliers and logistics demanders. The following example illustrates the normalization process using the expected evaluation values χ_a^s provided by the logistics supplier.

In Eq. (12), with respect to an evaluation value characterized by the HFLTS $H_a^s = \{o_a^{s\kappa} | \kappa = 1, 2, \dots, \#H_a^s\}$ and the given domain $[X_{\min}, X_{\max}] = [0, 1]$, H_a^s can be transformed into a cloud model by Eqs. (3)–(11). This model is denoted as $z_{H_a^s} = ([Ex_{H_a^s}, \bar{Ex}_{H_a^s}], En_{H_a^s}, He_{H_a^s})$. For an evaluation value represented by a triangular fuzzy number $k_a^s = (\underline{c}_a^s, c_a^s, \bar{c}_a^s)$, where the upper and lower bounds may be greater than 1, normalization is necessary. The normalization formula for a triangular fuzzy number is given as follows:

$$\tilde{k}_a^s = \begin{cases} (\underline{c}_a^s / \bar{c}_{\max a}^s, c_a^s / \bar{c}_{\max a}^s, \bar{c}_a^s / \bar{c}_{\max a}^s), & k_a^s \in \Lambda^c \\ [1 - (\underline{c}_a^s / \bar{c}_{\max a}^s), 1 - (c_a^s / \bar{c}_{\max a}^s), 1 - (\bar{c}_a^s / \bar{c}_{\max a}^s)], & k_a^s \in \Lambda^d \end{cases} \quad (16)$$

where $\bar{c}_{\max a}^s = \max\{\bar{c}_a^s | a = 1, 2, \dots, p\}$. For an evaluation value represented by a real number $y_a^s = \rho_a^s$, where the upper and lower bounds may be greater than 1, normalization is necessary. The normalization formula for a real number is given as follows:

$$\tilde{y}_a^s = \begin{cases} \rho_a^s / \rho_{\max a}^s, & y_a^s \in \Lambda^c \\ 1 - (\rho_a^s / \rho_{\max a}^s), & y_a^s \in \Lambda^d \end{cases} \quad (17)$$

where $\rho_{\max a}^s = \max\{\rho_a^s | a = 1, 2, \dots, p\}$. For the evaluation value characterized by the IFN $\vartheta_b^f = \langle \mu_b^f, \nu_b^f \rangle$, no normalization is needed. Based on the analysis above, if the normalized

expected evaluation value is denoted as $\tilde{\chi}_a^s$, then $\tilde{\chi}_a^s$ can be expressed as follows:

$$\tilde{\chi}_a^s = \begin{cases} z_{\tilde{H}_a^s} = ([Ex_{\tilde{H}_a^s}, \overline{Ex}_{\tilde{H}_a^s}], En_{\tilde{H}_a^s}, He_{\tilde{H}_a^s}) & \text{if } M_s \in \Lambda^1 \\ \tilde{y}_a^s = \tilde{\rho}_a^s & \text{if } M_s \in \Lambda^2 \\ \tilde{k}_a^s = (\tilde{c}_a^s, \tilde{c}_a^s, \tilde{c}_a^s) & \text{if } M_s \in \Lambda^3 \\ \tilde{\vartheta}_a^s = \langle \tilde{\mu}_a^s, \tilde{\nu}_a^s \rangle & \text{if } M_s \in \Lambda^4 \end{cases} \quad (18)$$

Similarly, if the normalized evaluation values γ_b^s , χ_a^f and γ_b^f are denoted as $\tilde{\gamma}_b^s$, $\tilde{\chi}_a^f$ and $\tilde{\gamma}_b^f$, then $\tilde{\gamma}_b^s$, $\tilde{\chi}_a^f$ and $\tilde{\gamma}_b^f$ can be represented as follows:

$$\tilde{\gamma}_b^s = \begin{cases} z_{\tilde{H}_b^s} = ([Ex_{\tilde{H}_b^s}, \overline{Ex}_{\tilde{H}_b^s}], En_{\tilde{H}_b^s}, He_{\tilde{H}_b^s}) & \text{if } M_s \in \Lambda^1 \\ \tilde{y}_b^s = \tilde{\rho}_b^s & \text{if } M_s \in \Lambda^2 \\ \tilde{k}_b^s = (\tilde{c}_b^s, \tilde{c}_b^s, \tilde{c}_b^s) & \text{if } M_s \in \Lambda^3 \\ \tilde{\vartheta}_b^s = \langle \tilde{\mu}_b^s, \tilde{\nu}_b^s \rangle & \text{if } M_s \in \Lambda^4 \end{cases} \quad (19)$$

$$\tilde{\chi}_a^f = \begin{cases} z_{\tilde{H}_a^f} = ([Ex_{\tilde{H}_a^f}, \overline{Ex}_{\tilde{H}_a^f}], En_{\tilde{H}_a^f}, He_{\tilde{H}_a^f}) & \text{if } N_f \in \Upsilon^1 \\ \tilde{y}_a^f = \tilde{\rho}_a^f & \text{if } N_f \in \Upsilon^2 \\ \tilde{k}_a^f = (\tilde{c}_a^f, \tilde{c}_a^f, \tilde{c}_a^f) & \text{if } N_f \in \Upsilon^3 \\ \tilde{\vartheta}_a^f = \langle \tilde{\mu}_a^f, \tilde{\nu}_a^f \rangle & \text{if } N_f \in \Upsilon^4 \end{cases} \quad (20)$$

$$\tilde{\gamma}_b^f = \begin{cases} z_{\tilde{H}_b^f} = ([Ex_{\tilde{H}_b^f}, \overline{Ex}_{\tilde{H}_b^f}], En_{\tilde{H}_b^f}, He_{\tilde{H}_b^f}) & \text{if } N_f \in \Upsilon^1 \\ \tilde{y}_b^f = \tilde{\rho}_b^f & \text{if } N_f \in \Upsilon^2 \\ \tilde{k}_b^f = (\tilde{c}_b^f, \tilde{c}_b^f, \tilde{c}_b^f) & \text{if } N_f \in \Upsilon^3 \\ \tilde{\vartheta}_b^f = \langle \tilde{\mu}_b^f, \tilde{\nu}_b^f \rangle & \text{if } N_f \in \Upsilon^4 \end{cases} \quad (21)$$

5.2 Computation of attribute weights

Considering the objectivity and subjectivity of the TSM decision process, a combined weight calculation method based on the BWM [45] and maximum deviation method [46] can be developed. The solution algorithm is as follows:

Algorithm 1:

Step 1.1: Considering the evaluation attribute set $M = \{M_1, M_2, \dots, M_s, \dots, M_e\}$, the optimal attribute M_B and the worst attribute M_W are determined.

Step 1.2: By using a 1–9 scale, the importance of the optimal attribute M_B relative to the other attribute M_s is measured, $s = 1, 2, \dots, e$, and a comparison vector $E_B = \{\varpi_{B1}, \varpi_{B2}, \dots, \varpi_{Bs}, \dots, \varpi_{Be}\}$ is constructed.

Step 1.3: By using a 1–9 scale, the importance of the worst attribute M_W relative to the other attribute M_s is measured, $s = 1, 2, \dots, e$, and a comparison vector $E_W = \{\varpi_{1W}, \varpi_{2W}, \dots, \varpi_{sW}, \dots, \varpi_{eW}\}$ is constructed.

Step 1.4: The model (M-1) is constructed based on comparison vectors $E_B = \{\varpi_{B1}, \varpi_{B2}, \dots, \varpi_{Bs}, \dots, \varpi_{Be}\}$ and $E_W = \{\varpi_{1W}, \varpi_{2W}, \dots, \varpi_{sW}, \dots, \varpi_{eW}\}$ as follows:

$$(M-1) \begin{cases} \text{Min } \delta \\ \text{s.t. } \left| \frac{w_B}{w_s} - E_{Bs} \right| \geq \delta \\ \left| \frac{w_s}{w_W} - E_{sW} \right| \geq \delta \\ \sum_{s=1}^e w_s^C = 1 \\ 0 \leq w_s^C \leq 1 \end{cases}$$

and the attribute weight set $w^C = \{w_1^C, w_2^C, \dots, w_s^C, \dots, w_e^C\}$ is obtained by solving the Model (M-1).

Step 1.5: The weighted Hamming distance between $\hat{\gamma}_b^s$ and the other actual evaluation value $\hat{\gamma}_{b'}^s$ under index M_s is computed as follows:

$$d_{bb'}^s(w_s^D) = \sum_{b'=1}^q d(\hat{\gamma}_b^s, \hat{\gamma}_{b'}^s) w_s^D \quad (22)$$

where

$$d(\hat{\gamma}_b^s, \hat{\gamma}_{b'}^s) = \begin{cases} \sqrt{\frac{1}{4} \left[\left(\overline{Ex}_{\tilde{H}_{b'}}^s - \overline{Ex}_{\tilde{H}_b}^s \right)^2 + \left(\overline{Ex}_{\tilde{H}_{b'}}^s - \overline{Ex}_{\tilde{H}_b}^s \right)^2 \right]} \\ \sqrt{\left(\tilde{\rho}_{b'}^s - \tilde{\rho}_b^s \right)^2} \\ \sqrt{\frac{1}{3} \left[\left(\tilde{c}_{b'}^s - \tilde{c}_b^s \right)^2 - \left(\tilde{c}_{b'}^s - \tilde{c}_b^s \right)^2 - \left(\tilde{c}_{b'}^s - \tilde{c}_b^s \right)^2 \right]} \\ \sqrt{\frac{1}{3} \left[\left(\tilde{\mu}_{b'}^s - \tilde{\mu}_b^s \right)^2 - \left(\tilde{v}_{b'}^s - \tilde{v}_b^s \right)^2 - \left(\tilde{\pi}_{b'}^s - \tilde{\pi}_b^s \right)^2 \right]} \end{cases} \quad (23)$$

Step 1.6: The total deviation of the evaluation index values with respect to the actual evaluation value is calculated as follows:

$$d^s = \sum_{b=1}^q d_b^s(w^D) = \sum_{b=1}^q \sum_{b'=1}^q d(\hat{\gamma}_b^s, \hat{\gamma}_{b'}^s) w_s^D \quad (24)$$

Step 1.7: Model (M-2) is constructed with the objective of maximizing the total deviation as follows:

$$(M-2) \begin{cases} \max d^s(w^D) = \sum_{s=1}^c \sum_{b=1}^z \sum_{b'=1}^z d(\hat{\gamma}_b^s, \hat{\gamma}_{b'}^s) w_s^D \\ \text{s.t. } \sum_{s=1}^e (w_s^D)^2 = 1, w_s^D \geq 1 \end{cases}$$

Step 1.8: Model (M-2) is solved by using the Lagrange multiplier method and normalized to obtain objective attribute weights as follows:

$$w_s^D = \frac{\sum_{b=1}^q \sum_{b'=1}^q d(\hat{\gamma}_b^s, \hat{\gamma}_{b'}^s)}{\sum_{s=1}^e \sum_{b=1}^q \sum_{b'=1}^q d(\hat{\gamma}_b^s, \hat{\gamma}_{b'}^s)} \quad (25)$$

Step 1.9: The weights of evaluation attributes are obtained by using a simple linear weighting approach to combine the subjective and objective weights derived from the BWM and the maximum deviation method as follows:

$$w_s = \lambda w_s^C + (1 - \lambda) w_s^D \quad (26)$$

where λ refers to the weight balance coefficient. If $\lambda = 0$, the decision-maker focuses only on the subjectivity of the decision process; if $\lambda = 1$, the decision-maker focuses only on the objectivity of the decision process.

Algorithm 2

Step 2.1: According to the evaluation attribute set $N = \{N_1, N_2, \dots, N_f, \dots, N_u\}$, the optimal attribute N_B and the worst attribute N_W are determined.

Step 2.2: By using a 1–9 scale, the importance of the optimal attribute N_B relative to the other attribute N_f is measured, $f = 1, 2, \dots, u$, and a comparison vector $E_B = \{\kappa_{B1}, \kappa_{B2}, \dots, \kappa_{Bf}, \dots, \kappa_{Bu}\}$ is constructed.

Step 2.3: By using a 1–9 scale, the importance of the worst attribute N_W relative to the other attribute N_f is measured, $f = 1, 2, \dots, u$, and a comparison vector $E_W = \{\kappa_{1W}, \kappa_{2W}, \dots, \kappa_{fW}, \dots, \kappa_{uW}\}$ is constructed.

Step 2.4: Model (M-3) is constructed based on comparison vectors $E_B = \{\kappa_{B1}, \kappa_{B2}, \dots, \kappa_{Bf}, \dots, \kappa_{Bu}\}$ and $E_W = \{\kappa_{1W}, \kappa_{2W}, \dots, \kappa_{fW}, \dots, \kappa_{uW}\}$ as follows:

$$(M-3) \begin{cases} \text{Min } \delta \\ \text{s.t. } \left| \frac{\omega_B}{\omega_f} - E_{Bf} \right| \geq \delta \\ \left| \frac{\omega_f}{\omega_W} - E_{fW} \right| \geq \delta \\ \sum_{f=1}^u \omega_f^C = 1, f = 1, 2, \dots, u \\ 0 \leq \omega_f^C \leq 1 \end{cases}$$

And the attribute weight set $\omega^C = \{\omega_1^C, \omega_2^C, \dots, \omega_f^C, \dots, \omega_u^C\}$ is obtained by solving the Model (M-3).

Step 2.5: The weighted Hamming distance between $\tilde{\chi}_a^f$ and the other actual evaluation value $\tilde{\chi}_{a'}^f$ under index N_f is computed as follows:

$$d_{aa'}^f(w_f^D) = \sum_{a'=1}^p d(\tilde{\chi}_a^f, \tilde{\chi}_{a'}^f) w_a^D \quad (27)$$

where

$$d(\tilde{\chi}_a^f, \tilde{\chi}_{a'}^f) = \begin{cases} \sqrt{\frac{1}{4} \left[\left(\overline{Ex}_{\tilde{H}_{a'}}^f - \overline{Ex}_{\tilde{H}_a}^f \right)^2 + \left(\overline{Ex}_{\tilde{H}_{a'}}^f - \overline{Ex}_{\tilde{H}_a}^f \right)^2 + \right.} \\ \left. \left(\overline{En}_{\tilde{H}_{a'}}^f - \overline{En}_{\tilde{H}_a}^f \right)^2 + \left(\overline{He}_{\tilde{H}_{a'}}^f - \overline{He}_{\tilde{H}_a}^f \right)^2 \right]} \\ \sqrt{\left(\tilde{\rho}_{a'}^f - \tilde{\rho}_a^f \right)^2} \\ \sqrt{\frac{1}{3} \left[\left(\tilde{c}_{a'}^f - \tilde{c}_a^f \right)^2 - \left(\tilde{c}_{a'}^f - \tilde{c}_a^f \right)^2 - \left(\tilde{c}_{a'}^f - \tilde{c}_a^f \right)^2 \right]} \\ \sqrt{\frac{1}{3} \left[\left(\tilde{\mu}_{a'}^f - \tilde{\mu}_a^f \right)^2 - \left(\tilde{v}_{a'}^f - \tilde{v}_a^f \right)^2 - \left(\tilde{\pi}_{a'}^f - \tilde{\pi}_a^f \right)^2 \right]} \end{cases} \quad (28)$$

Step 2.6: The total deviation of all the evaluation index values with respect to the actual evaluation value is calculated as follows:

$$d^f = \sum_{a=1}^p d_a^f(\omega^D) = \sum_{a=1}^q \sum_{a'=1}^q d(\tilde{\chi}_a^f, \tilde{\chi}_{a'}^f) \omega_f^D \quad (29)$$

Step 2.7: Model (M-4) is constructed with the objective of maximizing the total deviation as follows:

$$(M-4) \begin{cases} \max d^f(\omega^D) = \sum_{s=1}^e \sum_{b=1}^z \sum_{b'=1}^z d(\tilde{\chi}_a^f, \tilde{\chi}_{a'}^f) \omega_f^D \\ \text{s.t. } \sum_{f=1}^u (\omega_f^D)^2 = 1, \omega_f^D \geq 1 \end{cases}$$

Step 2.8: Model (M-4) is solved using the Lagrange multiplier method and normalized to obtain objective attribute weights as follows:

$$\omega_f^D = \frac{\sum_{a=1}^p \sum_{a'=1}^p d(\tilde{\chi}_a^f, \tilde{\chi}_{a'}^f)}{\sum_{f=1}^u \sum_{a=1}^p \sum_{a'=1}^p d(\tilde{\chi}_a^f, \tilde{\chi}_{a'}^f)} \quad (30)$$

Step 2.9: The weights of evaluation attributes are obtained by using a simple linear weighting approach to combine the subjective and objective weights derived from the BWM and the maximum deviation method as follows:

$$\omega_f = \lambda \omega_f^C + (1 - \lambda) \omega_f^D \quad (31)$$

where λ refers to the weight balance coefficient. If $\lambda = 0$, the decision-maker focuses only on the subjectivity of the decision process; if $\lambda = 1$, the decision-maker focuses only on the objectivity of the decision process.

5.3 Calculation of matching utility values and construction of the expected matching matrix

First, the Hamming distance measure between the normalized actual evaluation value $\tilde{x}_a^s$ and expected value $\tilde{y}_b^s$ can be calculated as follows:

$$d(\tilde{x}_a^s, \tilde{y}_b^s) = \begin{cases} \sqrt{\frac{1}{4} \left[\left(\overline{Ex}_{\tilde{H}_a^s} - \overline{Ex}_{\tilde{H}_b^s} \right)^2 + \left(\overline{Ex}_{\tilde{H}_a^s} - \overline{Ex}_{\tilde{H}_b^s} \right)^2 + \right.} \\ \left. \left(\overline{En}_{\tilde{H}_a^s} - \overline{En}_{\tilde{H}_b^s} \right)^2 + \left(\overline{He}_{\tilde{H}_a^s} - \overline{He}_{\tilde{H}_b^s} \right)^2 \right]} \\ \sqrt{(\tilde{\rho}_a^s - \tilde{\rho}_b^s)^2} \\ \sqrt{\frac{1}{3} \left[(\tilde{c}_a^s - \tilde{c}_b^s)^2 - (\tilde{c}_a^s - \tilde{c}_b^s)^2 - (\tilde{c}_a^s - \tilde{c}_b^s)^2 \right]} \\ \sqrt{\frac{1}{3} \left[(\tilde{\mu}_a^s - \tilde{\mu}_b^s)^2 - (\tilde{\nu}_a^s - \tilde{\nu}_b^s)^2 - (\tilde{\pi}_a^s - \tilde{\pi}_b^s)^2 \right]} \end{cases} \quad (32)$$

On this basis, the benefit-loss value η_{ab}^s of the logistics supplier with respect to the logistics demander under the attribute can be calculated as follows:

$$\eta_{ab}^s = \begin{cases} d(\tilde{x}_a^s, \tilde{y}_b^s), & \tilde{x}_a^s \geq \tilde{y}_b^s, \\ -d(\tilde{x}_a^s, \tilde{y}_b^s), & \tilde{x}_a^s < \tilde{y}_b^s \end{cases} \quad (33)$$

Then, the prospect value ς_{ab}^s of the logistics supplier with respect to the logistics demander under the attribute can be determined as follows:

$$\varsigma_{ab}^s = \begin{cases} \sqrt{\eta_{ab}^s}, & \tilde{x}_a^s \geq \tilde{y}_b^s \\ -\theta \sqrt{-\eta_{ab}^s}, & \tilde{x}_a^s < \tilde{y}_b^s \end{cases} \quad (34)$$

where θ refers to the decision-maker's sensitivity to loss. The larger the loss aversion coefficient is, the greater the disparity between the decision-maker's sensitivity to losses and gains. Conversely, a smaller coefficient indicates a smaller disparity between sensitivity to losses and gains.

Using Eqs. (32–34), the matching utility value σ_{ab} of the logistics supplier with respect to the logistics demander can be computed as follows:

$$\sigma_{ab} = \frac{\sum_{s=1}^v \varsigma_{ab}^s w^s - \min\{\sum_{s=1}^v \varsigma_{ab}^s w^s\}}{\max\{\sum_{s=1}^v \varsigma_{ab}^s w^s\} - \min\{\sum_{s=1}^v \varsigma_{ab}^s w^s\}} \quad (35)$$

According to the matching utility value, we can construct a unilateral matching Model (M-5) for logistics suppliers to logistics demanders as follows:

$$(M-5) \begin{cases} \text{Max } G_1 = \sum_{a=1}^p \sum_{b=1}^q \sigma_{ab} x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} \leq n, \quad b = 1, 2, \dots, q \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, p, \quad b = 1, 2, \dots, q \end{cases}$$

where $\sum_{a=1}^p x_{ab} \leq n$ represents that each logistics supplier can be matched with less than n logistics demanders, $n \leq q$; n refers to the initial expected number of logistics demanders to be matched for a logistics supplier.

Similarly, the Hamming distance measure between the normalized actual evaluation value $\tilde{\chi}_a^f$ and the expected value $\tilde{\gamma}_b^f$ can be calculated as follows.

$$d(\tilde{\chi}_a^f, \tilde{\gamma}_b^f) = \begin{cases} \sqrt{\frac{1}{4} \left[\left(\overline{Ex}_{\tilde{H}_a^f} - \overline{Ex}_{\tilde{H}_b^f} \right)^2 + \left(\overline{Ex}_{\tilde{H}_a^f} - \overline{Ex}_{\tilde{H}_b^f} \right)^2 \right]} \\ \sqrt{\left(\tilde{\rho}_a^f - \tilde{\rho}_b^f \right)^2} \\ \sqrt{\frac{1}{3} \left[\left(\tilde{c}_a^f - \tilde{c}_b^f \right)^2 - \left(\tilde{c}_a^f - \tilde{c}_b^f \right)^2 - \left(\tilde{c}_a^f - \tilde{c}_b^f \right)^2 \right]} \\ \sqrt{\frac{1}{3} \left[\left(\tilde{\mu}_a^f - \tilde{\mu}_b^f \right)^2 - \left(\tilde{v}_a^f - \tilde{v}_b^f \right)^2 - \left(\tilde{\pi}_a^f - \tilde{\pi}_b^f \right)^2 \right]} \end{cases} \quad (36)$$

On this basis, the benefit-loss value κ_{ab}^f of the logistics demander with respect to the logistics supplier under the attribute can be computed as follows:

$$\kappa_{ab}^f = \begin{cases} d(\tilde{\chi}_a^f, \tilde{\gamma}_b^f), & \tilde{\chi}_a^f \geq \tilde{\gamma}_b^f \\ -d(\tilde{\chi}_a^f, \tilde{\gamma}_b^f), & \tilde{\chi}_a^f < \tilde{\gamma}_b^f \end{cases} \quad (37)$$

Then, the prospect value v_{ab}^f of the logistics demander with respect to the logistics supplier under the attribute can be represented as follows:

$$v_{ab}^f = \begin{cases} \sqrt{\tilde{\kappa}_{ab}^f}, & \tilde{\chi}_a^f \geq \tilde{\gamma}_b^f \\ -\theta \sqrt{-\tilde{\kappa}_{ab}^f}, & \tilde{\chi}_a^f < \tilde{\gamma}_b^f \end{cases} \quad (38)$$

According to Eqs. (36–38), the matching utility value o_{ab} of the logistics demander with respect to the logistics supplier can be computed as follows:

$$o_{ab} = \frac{\sum_{f=1}^t o_{ab}^f \omega^f - \min\{\sum_{f=1}^t o_{ab}^f \omega^f\}}{\max\{\sum_{f=1}^t o_{ab}^f \omega^f\} - \min\{\sum_{f=1}^t o_{ab}^f \omega^f\}} \quad (39)$$

According to the matching utility value, a unilateral matching Model (M-6) for logistics suppliers to logistics demanders can be constructed, i.e.,

$$(M-6) \begin{cases} \text{Max } G_2 = \sum_{\zeta=1}^e \sum_{b=1}^z o_{ab} x_{ab} \\ \text{s.t. } \sum_{b=1}^q x_{ab} \leq m, \quad a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, p, \quad b = 1, 2, \dots, q \end{cases}$$

where $\sum_{b=1}^q x_{ab} \leq m$ represents that each logistics demander can be matched with less than m logistics suppliers, $m \leq p$; m refers to the initial expected number of logistics demanders to be matched for a logistics supplier.

Based on the above analysis, a new algorithm for obtaining the expected matching matrix is as follows:

Algorithm 3

Step 3.1: The matching utility σ_{ab} is calculated via Eq. (32–35).

Step 3.2: A one-sided matching Model (M-5) for logistics suppliers to logistics demanders is constructed based on matching utility values σ_{ab} ($a = 1, 2, \dots, p, b = 1, 2, \dots, q$).

Step 3.3: Model (M-5) is solved, and the candidate matching pairs x_{ab}^α ($a = 1, 2, \dots, p, b = 1, 2, \dots, q$) are obtained.

Step 3.4: For the candidate matching pairs x_{ab}^α , retain the matching utility value corresponding to the logistics demander that the logistics suppliers expect to match with is retained; the matching utility values for the remaining logistics demanders are set as 0. This yields in the expected matching matrix $PER_{p \times q}^{R \rightarrow T}$.

Step 3.5: The matching utility values σ_{ab} are calculated via Eqs. (36–39).

Step 3.6: A unilateral matching Model (M-6) for logistics demanders to logistics suppliers is constructed based on the matching utility values σ_{ab} ($a = 1, 2, \dots, p, b = 1, 2, \dots, q$).

Step 3.7: Model (M-6) is solved, and the candidate matching pairs x_{ab}^β ($a = 1, 2, \dots, p, b = 1, 2, \dots, q$) are obtained.

Step 3.8: For the candidate matching pairs x_{ab}^β , the matching utility value corresponding to the logistics supplier that the logistics demanders expect to match with is retained; the matching utility values for the remaining logistics suppliers are set to 0. This yields in the expected matching matrix $PER_{p \times q}^{T \rightarrow R}$.

5.4 Construction of the one-to-one stable matching model

According to the expected matching matrices $PER_{p \times q}^{R \rightarrow T}$ and $PER_{p \times q}^{T \rightarrow R}$ considering the regret aversion psychology of logistics suppliers and logistics demanders in the second matching process, the following steps are taken to calculate agent satisfaction based on regret theory.

The regret value $Y_{abj}^{R \rightarrow T}$ of matching logistics supplier R_a with logistics demander T_b compared to logistics demander T_j can be calculated as follows:

$$Y_{abj}^{R \rightarrow T} = \begin{cases} 1 - \exp[-\vartheta(\sigma_{ab} - \sigma_{aj})] & \sigma_{ab} < \sigma_{aj} \\ 0 & \sigma_{ab} \geq \sigma_{aj} \end{cases} \quad (40)$$

Then, the joy value $Q_{abj}^{R \rightarrow T}$ of matching logistics supplier R_a with logistics demander T_b compared to logistics demander T_j can be calculated as follows:

$$Q_{abj}^{R \rightarrow T} = \begin{cases} 0 & \sigma_{ab} < \sigma_{aj} \\ 1 - \exp[-\vartheta(\sigma_{ab} - \sigma_{aj})] & \sigma_{ab} \geq \sigma_{aj} \end{cases} \quad (41)$$

Hence, the relative satisfaction degree $O_{ab}^{R \rightarrow T}$ of matching logistics supplier R_a with logistics demander T_b compared to all other logistics demanders is calculated as follows:

$$O_{ab}^{R \rightarrow T} = \sum_{j=1}^q Y_{abj}^{R \rightarrow T} + \sum_{j=1}^q Q_{abj}^{R \rightarrow T} \quad (42)$$

By Eqs. (40–42), the agent satisfaction of the logistics supplier R_a for the logistics demander T_b can be calculated as follows:

$$W_{ab}^{R \rightarrow T} = \frac{O_{ab}^{R \rightarrow T} - \min\{O_{ab}^{R \rightarrow T}\}}{\max\{O_{ab}^{R \rightarrow T}\} - \min\{O_{ab}^{R \rightarrow T}\}} \quad (43)$$

Similarly, the regret $Y_{abu}^{T \rightarrow R}$ of matching logistics demander T_b with logistics supplier R_a compared to logistics supplier R_u can be calculated as follows:

$$Y_{abu}^{T \rightarrow R} = \begin{cases} 1 - \exp[-\vartheta(o_{ab} - o_{ub})] & o_{ab} < o_{ub} \\ 0 & o_{ab} \geq o_{ub} \end{cases} \quad (44)$$

The joy value $Q_{abu}^{T \rightarrow R}$ of matching logistics demander T_b with logistics supplier R_a compared to logistics supplier R_u is calculated as follows:

$$Q_{abu}^{T \rightarrow R} = \begin{cases} 0 & o_{ab} < o_{ub} \\ 1 - \exp[-\vartheta(o_{ab} - o_{ub})] & o_{ab} \geq o_{ub} \end{cases} \quad (45)$$

Hence, the relative satisfaction degree $O_{ab}^{T \rightarrow R}$ of matching logistics demander T_b with logistics supplier R_a compared to all other logistics suppliers is calculated as follows:

$$O_{ab}^{T \rightarrow R} = \sum_{u=1}^p Y_{abu}^{T \rightarrow R} + \sum_{u=1}^p Q_{abu}^{T \rightarrow R} \quad (46)$$

By Eqs. (44–46), the agent satisfaction of the logistics demander T_b for the logistics supplier R_a can be calculated as follows:

$$W_{ab}^{T \rightarrow R} = \frac{O_{ab}^{T \rightarrow R} - \min\{O_{ab}^{T \rightarrow R}\}}{\max\{O_{ab}^{T \rightarrow R}\} - \min\{O_{ab}^{T \rightarrow R}\}} \quad (47)$$

Furthermore, according to reference [18], it is known that if there is a successful match between the logistics supplier and the logistics demander, and the logistics supplier has the highest satisfaction with the logistics demander, then a higher fee should be paid. Conversely, if the logistics supplier has the lowest satisfaction with the logistics demander, the fee paid to the logistics O2O platform would also be the lowest. Based on this idea, the definition of expected the intermediary benefit is given below.

Definition 10 [18] Let ξ_{ab} be the expected intermediary benefit that the logistics supplier pays to the logistics O2O platform when successfully matches with the logistics demander and ψ_{ab} be expected the intermediary benefit that the logistics demander pays to the logistics O2O platform when successfully matches with the logistics supplier, then ξ_{ab} and ψ_{ab} are presented as follows:

$$\xi_{ab} = \left[\frac{q + 1 + W_{ab}^{R \rightarrow T}}{q} \right]^2 \quad (48)$$

$$\psi_{ab} = \left[\frac{p + 1 + W_{ab}^{T \rightarrow R}}{p} \right]^2 \quad (49)$$

where ξ_{ab} and ψ_{ab} are monotonically increasing functions.

However, the expected intermediary benefit in Definition [8] does not account for influence of platform risk preference. In the decision process of TSM in real logistics O2O platforms, the willingness of TSM agents to match will to some extent determine whether the TSM scheme can be reached, and the intermediary's risk preference will also affect the TSM scheme. Therefore, a calculation approach for the expected intermediary benefit considering risk preference coefficients is proposed, and the risk preference coefficient is based on the willingness of agents to match.

Definition 11 Let $W_{ab}^{R \rightarrow T}$ be the satisfaction of logistics supplier R_a to logistics demander T_b and $W_{ab}^{T \rightarrow R}$ be the satisfaction of logistics demander T_b to the logistics supplier R_a . Let ε_{ab} be the matching willingness coefficient between logistics supplier R_a and logistics demander T_b . Then, expected the intermediary benefit ε_{ab} can be obtained as follows:

$$\varepsilon_{ab} = \frac{e^{\sqrt{W_{ab}^{R \rightarrow T} W_{ab}^{T \rightarrow R}}}}{1 + e^{|W_{ab}^{R \rightarrow T} - W_{ab}^{T \rightarrow R}|}} \quad (50)$$

Here, $\sqrt{W_{ab}^{R \rightarrow T} W_{ab}^{T \rightarrow R}}$ refers to the consistency among the agents, and $|W_{ab}^{R \rightarrow T} - W_{ab}^{T \rightarrow R}|$ refers to the deviation among the agents. The higher the consistency among the agents and the smaller the deviation between them are, the stronger the inclination for harmonious alignment between logistics supplier and logistics demander. Conversely, if there is a lower level of consistency among agents and a larger deviation between them, the inclination for alignment is weaker.

Definition 12 Let ξ_{ab} be the expected intermediary benefit that logistics supplier R_a pays to the logistics O2O platform when R_a is successfully matched with logistics demander T_b ; let ψ_{ab} be the expected intermediary benefit that logistics demander T_b pays to the logistics O2O platform when T_b is successfully matched with logistics supplier R_a . Then ξ_{ab} and ψ_{ab} can be obtained as follows:

$$\xi_{ab} = v \left(\frac{q + W_{ab}^{R \rightarrow T}}{p + q + 1} \right)^{\frac{1}{2}} \quad (51)$$

$$\psi_{ab} = v \left(\frac{p + W_{ab}^{T \rightarrow R}}{p + q + 1} \right)^{\frac{1}{2}} \quad (52)$$

Here, v represents the risk preference coefficient. If the platform is risk seeking, then $v = \min\{\varepsilon_{ab}\}$; if the platform is risk neutral, then $v = \text{mid}\{\varepsilon_{ab}\}$, where mid represents the median; if the platform is risk averse, then $v = \max\{\varepsilon_{ab}\}$.

5.5 Construction of the one-to-many matching model

Furthermore, according to agent satisfactions $W_{ab}^{R \rightarrow T}$ and $W_{ab}^{T \rightarrow R}$, and intermediary benefits ξ_{ab} and ψ_{ab} , the following TSM Model (M-7) can be constructed:

$$(M-7) \left\{ \begin{array}{l} \text{Max } Z_1 = \sum_{a=1}^p \sum_{b=1}^q W_{ab}^{R \rightarrow T} x_{ab} \\ \text{Max } Z_2 = \sum_{a=1}^p \sum_{b=1}^q W_{ab}^{T \rightarrow R} x_{ab} \\ \text{Max } Z_3 = \sum_{a=1}^p \sum_{b=1}^q (\xi_{ab} + \psi_{ab}) x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, \quad b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, \quad a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, p, \quad b = 1, 2, \dots, q \end{array} \right.$$

In Model (M-7), $\text{Max } Z_1 = \sum_{a=1}^p \sum_{b=1}^q W_{R \rightarrow T} x_{ab}$ denotes the maximization of the total satisfaction of logistics suppliers toward logistics demanders; $\text{Max } Z_2 = \sum_{a=1}^p \sum_{b=1}^q W_{T \rightarrow R} x_{ab}$ denotes the maximization of the total satisfaction of logistics demanders toward logistics suppliers; $\text{Max } Z_3 = \sum_{a=1}^p \sum_{b=1}^q (\xi_{ab} + \psi_{ab}) x_{ab}$ represents the maximization of the expected intermediary benefit; $\sum_{a=1}^p x_{ab} = \iota$ represents that the logistics supplier can be matched with less than ι logistics providers, where $2 \leq \iota \leq m$; $\sum_{b=1}^q x_{ab} \leq 1$ represents that the logistics demander can be matched with at most a logistics supplier; $x_{ab} = \{0, 1\}$ represents that 0–1 variable, where $x_{ab} = 0$ represents the logistics supplier and the logistics demander are not successfully matched, and $x_{ab} = 1$ represents that the logistics supplier and the logistics demander are successfully matched.

5.6 Solution of the one-to-many matching model

For the multiobjective programming Model (M-7), a new max–min solution algorithm that accounts for fairness is proposed as follows:

Algorithm 4

Step 4.1: The following Model (M-8) is solved, and its maximum value $Z_1^{\max}$ is calculated.

$$(M-8) \left\{ \begin{array}{l} \text{Max } Z_1 = \sum_{a=1}^p \sum_{b=1}^q W_{ab}^{R \rightarrow T} x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, \quad b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, \quad a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, p, b = 1, 2, \dots, q \end{array} \right.$$

Step 4.2: The following Model (M-9) is solved, and its minimum value $Z_1^{\min}$ is calculated.

$$(M-9) \left\{ \begin{array}{l} \text{Min } Z_1 = \sum_{a=1}^p \sum_{b=1}^q W_{ab}^{R \rightarrow T} x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, \quad b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, \quad a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, p, b = 1, 2, \dots, q \end{array} \right.$$

Step 4.3: The following Model (M-10) is solved, and its maximum value $Z_2^{\max}$ is calculated.

$$(M-10) \left\{ \begin{array}{l} \text{Max } Z_2 = \sum_{a=1}^p \sum_{b=1}^q W_{ab}^{T \rightarrow R} x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, \quad b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, \quad a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, p, b = 1, 2, \dots, q \end{array} \right.$$

Step 4.4: The following Model (M-11) is solved, and its minimum value $Z_2^{\min}$ is calculated.

$$(M-11) \left\{ \begin{array}{l} \text{Min } Z_2 = \sum_{a=1}^p \sum_{b=1}^q W_{ab}^{T \rightarrow R} x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, a = 1, 2, \dots, p, b = 1, 2, \dots, q \end{array} \right.$$

Step 4.5: The following Model (M-12) is solved, and its maximum value $Z_3^{\max}$ is calculated.

$$(M-12) \left\{ \begin{array}{l} \text{Max } Z_3 = \sum_{a=1}^p \sum_{b=1}^q (\xi_{ab} + \psi_{ab}) x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, a = 1, 2, \dots, p, b = 1, 2, \dots, q \end{array} \right.$$

Step 4.6: The following Model (M-13) is solved, and its minimum value $Z_3^{\min}$ is calculated.

$$(M-13) \left\{ \begin{array}{l} \text{Min } Z_3 = \sum_{a=1}^p \sum_{b=1}^q (\xi_{ab} + \psi_{ab}) x_{ab} \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, a = 1, 2, \dots, p, b = 1, 2, \dots, q \end{array} \right.$$

Step 4.7: A new membership function that incorporates the fairness idea is constructed as follows:

$$\begin{aligned} \mu_{Z_1} &= \frac{Z_1 - Z_1^{\min}}{\bar{Z}} \\ \mu_{Z_2} &= \frac{Z_2 - Z_2^{\min}}{\bar{Z}} \\ \mu_{Z_3} &= \frac{Z_3 - Z_3^{\min}}{\bar{Z}} \end{aligned}$$

where $\bar{Z} = (Z_1^{\max} - Z_1^{\min}) + (Z_2^{\max} - Z_2^{\min}) + (Z_3^{\max} - Z_3^{\min})$.

Step 4.8: The following single objective TSM Model (M-14) is constructed:

$$(M-14) \left\{ \begin{array}{l} \text{Max } \mu \\ 3\mu_{Z_1} + \mu_{Z_1} + \mu_{Z_2} + \mu_{Z_3} \geq 6\mu \\ 3\mu_{Z_2} + \mu_{Z_1} + \mu_{Z_2} + \mu_{Z_3} \geq 6\mu \\ 3\mu_{Z_3} + \mu_{Z_1} + \mu_{Z_2} + \mu_{Z_3} \geq 6\mu \\ \text{s.t. } \sum_{a=1}^p x_{ab} = \iota, \quad b = 1, 2, \dots, q \\ \sum_{b=1}^q x_{ab} \leq 1, \quad a = 1, 2, \dots, p \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, p, \quad b = 1, 2, \dots, q \end{array} \right.$$

Step 4.9: Model (M-14) is solved, and the optimal TSM scheme is obtained.

Remark: In the traditional membership weighting function $\mu_{Z_1} = \frac{Z_1^{\max} - Z_1}{Z_1^{\max} - Z_1^{\min}}$, only the difference in the objective Z_1 itself is considered. In the new membership degree weighting function $\mu_{Z_1} = \frac{Z_1 - Z_1^{\min}}{\bar{Z}}$, the difference between the objective Z_2 and objective Z_3 is added to the denominator. The fairness idea in the new membership degree weighting function incorporating the objective differences of other objective functions. There is a similar inference for objective functions Z_2 and Z_3 .

5.7 Solution process

In summary, the solution steps of one-to-many O2O supply–demand TSM decision method considering fairness are presented as follows:

Step 1: The multiple fuzzy preference information is normalized via Eqs. (1)–(21).

Step 2: The weight vectors $w = \{w_1, w_2, \dots, w_s, \dots, w_e\}$ and $\omega = \{\omega_1, \omega_2, \dots, \omega_f, \dots, \omega_u\}$ are determined by using Algorithm 1 and Algorithm 2, respectively.

Step 3: The matching utility values σ_{ab} and o_{ab} are calculated via Eqs. (32–39).

Step 4: The expected matching matrices $PER_{p \times q}^{R \rightarrow T}$ and $PER_{p \times q}^{T \rightarrow R}$ are determined by using Algorithm.

Step 5: Obtain the agent satisfactions $W_{ab}^{R \rightarrow T}$ and $W_{ab}^{T \rightarrow R}$ by Eqs. (40–47).

Step 6: A multi-objective programming model considering the intermediary benefit objective and one-to-many matching constraints based on agent satisfactions $W_{ab}^{R \rightarrow T}$ and $W_{ab}^{T \rightarrow R}$ is constructed.

Step 7: Model (M-7) is solved by using Algorithm 4 and a reasonable one-to-many TSM scheme on the logistics O2O platform is obtained.

6 Case study

6.1 Solution process

HLL Company is a logistics O2O enterprise in the Greater Bay Area of Guangdong, Hong Kong, and Macau. It operates in various aspects of the business, including local and cross-city freight services, less-than-truckload transportation, after-sales services for vehicles, and car rental services. Through the freight O2O platform developed by HLL, shippers with transportation needs can quickly find truck resources at a lower price, ensuring high-quality freight services. Likewise, trucks can efficiently connect with potential shippers with transport demands through the platform, thereby enhancing truck utilization efficiency. Eight shippers $\{T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8\}$ have sent orders through the platform due to their freight demands. At the same time, 4 truck drivers $\{R_1, R_2, R_3, R_4\}$ have received these orders. Due to the truck loading capacity and vehicle safety limitations, each truck can serve a maximum of two shippers. In the service evaluation system provided by the HLL Company, truck drivers $\{R_1, R_2, R_3, R_4\}$ evaluate shippers $\{T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8\}$ based on attributes such as breach of contract incidents, cancellation rate, positive feedback rate, business growth rate, and credit rating. Shippers $\{T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8\}$ evaluate truck drivers $\{R_1, R_2, R_3, R_4\}$ based on attributes such as completion volume, mental resilience, cargo damage rate, on-time delivery rate, and vehicle equipment status. The shippers $\{T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8\}$ and truck drivers $\{R_1, R_2, R_3, R_4\}$ provide their expected evaluation values based on their actual needs, as shown in Tables 2 and 3. The platform then generates the actual evaluation values based on the information provided by both sides, as shown in Tables 4 and 5. The expected evaluation values and actual evaluation values incorporate four types of evaluation information: HFLTS, triangular fuzzy numbers, real numbers and IFNs. Here, M_1 and N_1 are real numbers; M_2, M_3, N_3 and N_4 are triangular fuzzy numbers; M_4 and N_2 are HFLTS; M_5 and N_5 are IFNs.

The decision steps for the presented one-to-many matching method on a logistics O2O platform considering the intermediary benefit are as follows:

Step 1: The fuzzy preference evaluation information provided by shippers and truck drivers is standardized via Eqs. (1–21).

Step 2: The weight vectors $w = \{w_1, w_2, \dots, w_5\}$ and $\omega = \{\omega_1, \omega_2, \dots, \omega_5\}$ are determined by using Algorithm 1 and Algorithm 2, respectively, where the balancing coefficient $\lambda = 0.6$.

Step 3: The matching utility values σ_{ab} and ϕ_{ab} are calculated via Eqs. (32–39), where $\theta = 2.25$ according to literature [22], as shown in Tables 6 and 7.

Step 4: The expected matching matrices $PER_{p \times q}^{R \rightarrow T}$ and $PER_{p \times q}^{T \rightarrow R}$ are determined by using Algorithm 3, as shown in Tables 8 and 9.

Table 2 Expected evaluation values of drivers χ_a^s ($a = 1, 2, 3, 4; s = 1, 2, \dots, 5$)

	M_1	M_2	M_3	M_4	M_5
R_1	3	(0, 0.2, 0.4)	(0.3, 0.6, 0.7)	$\{S_{-1}, S_1\}$	$< 0.7, 0.3 >$
R_2	0	(0.2, 0.3, 0.5)	(0.2, 0.4, 0.7)	$\{S_{-2}, S_2\}$	$< 0.5, 0.2 >$
R_3	2	(0.4, 0.8, 1)	(0.1, 0.5, 1)	$\{S_0, S_2\}$	$< 0.9, 0.1 >$
R_4	1	(0.2, 0.4, 0.5)	(0.5, 0.6, 0.7)	$\{S_{-2}, S_1\}$	$< 0.4, 0.3 >$

Table 3 Expected evaluation values of shippers γ_b^f ($b = 1, 2, \dots, 8$; $f = 1, 2, \dots, 5$)

	N_1	N_2	N_3	N_4	N_5
T_1	134	$\{S_{-2}, S_0\}$	(0.3, 0.4, 0.9)	(0.5, 0.7, 1)	$< 0.1, 0.7 >$
T_2	167	$\{S_{-1}, S_0\}$	(0.1, 0.6, 0.7)	(0.3, 0.4, 0.9)	$< 0.3, 0.2 >$
T_3	132	$\{S_1, S_2\}$	(0.2, 0.3, 0.5)	(0, 0.3, 0.8)	$< 0.4, 0.1 >$
T_4	89	$\{S_0, S_1\}$	(0.2, 0.5, 1)	(0.4, 0.9, 1)	$< 0.6, 0.3 >$
T_5	113	$\{S_{-2}, S_{-1}\}$	(0, 0.4, 0.8)	(0.3, 0.6, 0.6)	$< 0.5, 0.5 >$
T_6	198	$\{S_1, S_2\}$	(0.2, 0.7, 0.9)	(0, 0.6, 0.7)	$< 0.8, 0.1 >$
T_7	172	$\{S_{-1}, S_1\}$	(0.1, 0.2, 0.4)	(0.3, 0.5, 0.8)	$< 0, 0.3 >$
T_8	200	$\{S_0, S_2\}$	(0, 0.7, 0.8)	(0.2, 0.7, 0.9)	$< 0.4, 0.2 >$

Table 4 Actual evaluation values γ_b^s ($b = 1, 2, \dots, 8$; $s = 1, 2, \dots, 5$) of shippers provided by the platform

	M_1	M_2	M_3	M_4	M_5
T_1	0	(0.4, 0.6, 0.6)	(0.1, 0.2, 0.3)	$\{S_{-2}, S_{-1}\}$	$< 0.4, 0.5 >$
T_2	4	(0.1, 0.3, 0.4)	(0.5, 0.8, 0.9)	$\{S_{-2}, S_1\}$	$< 0.1, 0.6 >$
T_3	9	(0.3, 0.5, 0.7)	(0.7, 0.8, 0.9)	$\{S_{-1}, S_0\}$	$< 0.6, 0.1 >$
T_4	5	(0.2, 0.3, 0.6)	(0.2, 0.5, 1)	$\{S_1, S_2\}$	$< 0.3, 0.4 >$
T_5	1	(0.1, 0.2, 0.6)	(0.3, 0.6, 0.7)	$\{S_{-2}, S_0\}$	$< 0.2, 0.3 >$
T_6	8	(0.3, 0.6, 0.7)	(0.2, 0.4, 0.7)	$\{S_1, S_2\}$	$< 0.3, 0.3 >$
T_7	10	(0.1, 0.2, 1)	(0.3, 0.5, 0.9)	$\{S_0, S_1\}$	$< 0.8, 0.1 >$
T_8	6	(0.3, 0.5, 0.9)	(0.3, 0.4, 0.6)	$\{S_{-2}, S_0\}$	$< 0.6, 0.3 >$

Table 5 Actual evaluation values χ_a^f ($a = 1, 2, 3, 4$; $f = 1, 2, \dots, 5$) of drivers provided by the platform

	N_1	N_2	N_3	N_4	N_5
R_1	187	$\{S_{-2}, S_0\}$	(0.2, 0.4, 0.5)	(0.5, 0.8, 0.9)	$< 0.5, 0.2 >$
R_2	165	$\{S_{-1}, S_0\}$	(0.1, 0.8, 1)	(0.2, 0.6, 0.7)	$< 0.7, 0.1 >$
R_3	160	$\{S_{-2}, S_0\}$	(0.2, 0.4, 0.6)	(0.5, 0.7, 1)	$< 0.6, 0.2 >$
R_4	200	$\{S_0, S_2\}$	(0, 0.2, 0.3)	(0.7, 0.8, 0.8)	$< 0.8, 0.1 >$

Table 6 Matching utility value σ_{ab}

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
R_1	0.9258	0.6595	0.4770	0.2443	0.6853	0.2050	0.5906	0.6128
R_2	0.4060	0.5387	0.5171	0.4635	0.8572	0.1359	0.2982	0.6900
R_3	0.7839	0.6739	0.6751	0.4956	0.8894	0.0000	0.6363	0.7029
R_4	1.0000	0.7228	0.5824	0.8879	0.7332	0.2932	0.6861	0.6552

Table 7 Matching utility value σ_{ab}

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
R_1	0.3923	0.7423	0.7999	0.5846	0.8422	0.4774	0.7691	0.4344
R_2	0.1860	0.4226	0.1119	0.3699	0.8117	0.0589	0.2066	0.0000
R_3	0.7230	0.9693	0.5795	0.9716	0.7953	0.5296	0.4023	0.9048
R_4	0.4026	0.7818	0.4570	0.3501	1.0000	0.0887	0.1261	0.1849

Table 8 Expected matching matrix $PER_{p \times q}^{R \rightarrow T}$

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
R_1	0.9258	0.6595	0	0	0.6853	0	0	0.6128
R_2	0	0.5387	0.5171	0	0.8572	0	0	0.6900
R_3	0.7839	0	0.6751	0	0.8894	0	0	0.7029
R_4	1.0000	0.7228	0	0.8879	0.7332	0	0	0

Table 9 Expected matching matrix $PER_{p \times q}^{T \rightarrow R}$

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
R_1	0	0	0.7999	0.5846	0.8422	0.4774	0.7691	0.4344
R_2	0	0	0	0	0	0	0	0
R_3	0.7230	0.9693	0.5795	0.9716	0	0.5296	0.4023	0.9048
R_4	0.4026	0.7818	0	0	1.0000	0	0	0

Step 5: Agent satisfactions $W_{ab}^{R \rightarrow T}$ and $W_{ab}^{T \rightarrow R}$ are determined by Eqs. (40–47), where $\vartheta = 0.3$ according to literature [29], as shown in Tables 10 and 11.

Step 6–7: Considering the platform as a risk-neutral type, a TSM Model (M-7) is constructed, and the optimal TSM scheme can be obtained by using Algorithm 4 to solve the Model (M-7), as shown in Table 12.

Table 10 Agent satisfaction $W_{ab}^{R \rightarrow T}$

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
R_1	0.9859	0.7470	0.0663	0.0663	0.7710	0.0663	0.0663	0.7031
R_2	0.1053	0.6653	0.6446	0.1053	0.9563	0.1053	0.1053	0.8070
R_3	0.8426	0.0430	0.7425	0.0430	0.9365	0.0430	0.0430	0.7684
R_4	1.0000	0.7521	0.0000	0.9022	0.7618	0.0000	0.0000	0.0000

Table 11 Agent satisfaction $W_{ab}^{T \rightarrow R}$

	T_1	T_2	T_3	T_4	T_5	T_6	T_7
R_1	0.0941	0.0125	0.3980	0.2949	0.3653	0.3135	0.4070
R_2	0.0941	0.0125	0.0622	0.0378	0.0000	0.1098	0.0880
R_3	0.3948	1.0000	0.3134	0.4420	0.0000	0.3285	0.2641
R_4	0.2695	0.3520	0.0622	0.0378	0.4241	0.1098	0.0880

Table 12 Optimal TSM scheme

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
R_1	0	0	0	1	0	0	1	0
R_2	0	0	1	0	0	1	0	0
R_3	0	1	0	0	0	0	0	1
R_4	1	0	0	0	1	0	0	0

It can be seen from Table 12 that is $\delta_1^* = \{(R_1, T_4), (R_1, T_7), (R_2, T_3), (R_2, T_6), (R_3, T_2), (R_3, T_8), (R_4, T_1), (R_4, T_5)\}$. Namely, R_1 is matched with T_4 and T_7 , R_2 is matched with T_3 and T_6 , R_3 is matched with T_2 and T_8 , and R_4 is matched with T_1 and T_5 .

6.2 Comparative analysis

First, the proposed TSM decision method comprehensively considers two types of psychological behavioral factors that matching agents may exhibit, namely, loss aversion and regret aversion. To verify the effectiveness of the proposed TSM method, Model (M-15) is redeveloped by using prospect theory to calculate the matching utility value, without considering

Table 13 Comparative analysis of whether to consider regret aversion psychology

Method	Optimal TSM scheme
TSM method considering regret theory (Method I)	$\delta_2^* = \{(R_1, T_3), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, R_8), (R_4, T_1), (R_4, T_2)\}$
TSM method without considering regret theory (Method II)	$\delta_3^* = \{(R_1, T_6), (R_1, T_7), (R_2, T_3), (R_2, T_5), (R_3, T_2), (R_3, T_8), (R_4, T_1), (R_4, T_4)\}$

regret aversion:

$$(M-15) \left\{ \begin{array}{l} \text{Max } Z_1 = \sum_{a=1}^4 \sum_{b=1}^8 \sigma_{ab} x_{ab} \\ \text{Max } Z_2 = \sum_{a=1}^4 \sum_{b=1}^8 o_{ab} x_{ab} \\ \text{Max } Z_3 = \sum_{a=1}^4 \sum_{b=1}^q (\xi_{ab} + \psi_{ab}) x_{ab} \\ \text{s.t. } \sum_{a=1}^4 x_{ab} = 2, \quad b = 1, 2, \dots, 8 \\ \sum_{b=1}^8 x_{ab} \leq 1, \quad a = 1, 2, \dots, 4 \\ x_{ab} = \{0, 1\}, \quad a = 1, 2, \dots, 4, b = 1, 2, \dots, 8 \end{array} \right.$$

Then, the optimal TSM scheme is obtained by using Algorithm 4 to solve Model (M-15). A comparative analysis on whether to consider regret aversion psychology is conducted on the optimal TSM schemes obtained in the two scenarios, as shown in Table 13.

Table 13 shows that there are significant differences between the optimal TSM schemes obtained by Method I and Method II. Therefore, the different psychological behavioral factors that exist in different matching decision stages have certain impacts on the optimal TSM scheme.

Second, the commonly used solution algorithms in the literature include the max–min solution method [15], linear weighting method [14], and membership function weighting method [36], and so on. Model (M-7) can be solved by using different algorithms; the comparative analysis of these algorithms is presented in Table 14.

Table 14 presents the matching results obtained using Algorithms 2–5, all of which yield the same optimal TSM scheme. However, the optimal TSM scheme obtained by Algorithm 1 differs significantly from those obtained by Algorithms 2–5. Compared with the above solution algorithms, the proposed Algorithm 1 not only optimizes each objective but also considers fairness among objectives.

Third, to demonstrate the effectiveness and reliability of the proposed TSM method, a comparative analysis is conducted on the TSM decision methods proposed in [12, 47, 48] and this paper, which is shown in Table 15.

Table 14 Comparative analysis of different solution algorithms

Solution algorithm	Optimal TSM scheme
Max–min algorithm considering fairness (Proposed Algorithm 1)	$\delta_2^* = \{(R_1, T_3), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$
Max–min algorithm (Algorithm 2)	$\delta_4^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_2), (R_4, T_3)\}$
Linear weighted algorithm (Algorithm 3)	$\delta_4^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_2), (R_4, T_3)\}$
Membership function weighted algorithm (Algorithm 4)	$\delta_4^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_2), (R_4, T_3)\}$
As optimal as possible algorithm (Algorithm 5)	$\delta_4^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_2), (R_4, T_3)\}$

Table 15 Comparative analysis of different TSM methods

Method	Optimal TSM scheme
Method proposed in literature [12]	$\delta_5^* = \{(R_1, T_1), (R_1, T_7), (R_2, T_5), (R_2, T_6), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$
Method proposed in literature [47]	$\delta_6^* = \{(R_1, T_3), (R_1, T_7), (R_2, T_5), (R_2, T_8), (R_3, T_2), (R_3, T_6), (R_4, T_1), (R_4, T_4)\}$
Method proposed in literature [48]	$\delta_7^* = \{(R_1, T_6), (R_1, T_7), (R_3, T_1), (R_3, T_3), (R_3, T_8), (R_4, T_2), (R_4, T_4), (R_4, T_5)\}$
Method proposed in this paper	$\delta_2^* = \{(R_1, T_3), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$

Table 16 reveals the variations in the optimal TSM schemes obtained by the aforementioned TSM decision methods. The proposed TSM method considers various psychological behavioral factors of two-sided agents, thus reflecting the real-world TSM decision behavior better than the TSM decision methods proposed in [12] and [47]. Second, compared with

Table 16 The similarity and difference of four results obtained by four methods

	δ_2^*	δ_5^*	δ_6^*	δ_7^*	Similarity	Difference
δ_2^*	–	5	3	3	11	13
δ_5^*	5	–	3	3	11	13
δ_6^*	3	3	–	2	8	16
δ_7^*	3	3	2	–	8	16

the TSM decision methods proposed in [12, 47] and [48], the TSM method proposed in this paper aims to optimize the satisfaction considering fairness. Throughout the model-solving process, fairness among two-sided agents is fully considered.

In addition, to further demonstrate the effectiveness of the proposed method and explain the similarity among the results obtained by these methods from literature [12, 47] and [48], Table 16 is provided. Table 16 intuitively presents the similarity and difference by comparing the numbers of the same and different matching pairs among the optimal schemes δ_2^* , δ_5^* , δ_6^* and δ_7^* . As shown in Table 16, it is reasonable to observe that both similarity and difference in the results obtained by different methods for the same problem with the same data. The method presented in this paper exhibits a high degree of similarity and a low degree of difference in the optimal scheme of matching pairs compared to other three methods, indicating that the proposed method is effective in achieving the intended results.

The risk aversion coefficient reflects the sensitivity of two-sided agents to loss. A higher risk aversion coefficient indicates a higher sensitivity to loss for the agent. Therefore, agents can choose different risk aversion coefficients based on their own preferences. In addition, this paper combines the BWM method is combined with maximum deviation method to calculate the weights of attributes, and introduces the weight balance coefficient. Agents can choose different weight balance coefficients based on their actual needs. To investigate the impacts of different risk aversion coefficients and weight balance coefficients on the optimal TSM scheme under different scenarios, five different risk aversion coefficients and weight balance coefficients are selected for each case. The obtained optimal TSM schemes under five situations are shown in Tables 17 and 18.

Table 17 shows that there are some differences in the obtained optimal TSM schemes considering different weight balance coefficients. When the weight balance coefficients are set to 0.1, 0.3, and 0.5, the optimal TSM schemes are the same for the three scenarios. However, when the weight balance coefficient is $\lambda = 0.7$ or $\lambda = 0.9$, the optimal TSM scheme varies. Therefore, it can be concluded that different weight balance coefficients can have certain impacts on the optimal TSM scheme. Additionally, Table 18 shows that there are significant differences in the obtained optimal TSM schemes when the risk aversion coefficients change in different scenarios. In conclusion, two-sided agents should choose different risk aversion coefficients and weight balance coefficients based on their own preferences and actual needs

Table 17 Optimal TSM scheme based on the weight balance coefficient ($\theta = 2.25$)

Weight balance coefficient	Optimal TSM scheme
$\lambda = 0.1$	$\delta_2^* = \{(R_1, T_3), (R_1, T_7), (R_2, T_5), (R_2, T_6), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$
$\lambda = 0.3$	$\delta_2^* = \{(R_1, T_3), (R_1, T_7), (R_2, T_5), (R_2, T_6), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$
$\lambda = 0.5$	$\delta_2^* = \{(R_1, T_3), (R_1, T_7), (R_2, T_5), (R_2, T_6), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$
$\lambda = 0.7$	$\delta_8^* = \{(R_1, T_4), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$
$\lambda = 0.9$	$\delta_9^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_8), (R_3, T_2), (R_3, T_3), (R_4, T_4), (R_4, T_7)\}$

Table 18 Optimal matching scheme based on the risk aversion coefficient ($\lambda = 0.6$)

Risk aversion coefficient	Optimal TSM scheme
$\theta = 0.45$	$\delta_{10}^* = \{(R_1, T_6), (R_1, T_7), (R_2, T_5), (R_2, T_8), (R_3, T_2), (R_3, T_3), (R_4, T_1), (R_4, T_4)\}$
$\theta = 0.9$	$\delta_9^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_8), (R_3, T_2), (R_3, T_3), (R_4, T_4), (R_4, T_7)\}$
$\theta = 1.35$	$\delta_9^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_8), (R_3, T_2), (R_3, T_3), (R_4, T_4), (R_4, T_7)\}$
$\theta = 1.8$	$\delta_9^* = \{(R_1, T_1), (R_1, T_6), (R_2, T_5), (R_2, T_8), (R_3, T_2), (R_3, T_3), (R_4, T_4), (R_4, T_7)\}$
$\theta = 2.25$	$\delta_2^* = \{(R_1, T_3), (R_1, T_6), (R_2, T_5), (R_2, T_7), (R_3, T_4), (R_3, T_8), (R_4, T_1), (R_4, T_2)\}$

when the method proposed in this paper is used to solve real-world TSM decision problems with multiple fuzzy preference information.

In summary, the method proposed in this paper is effective in addressing the supply–demand matching problem on the freight O2O platform developed by HLL, and has universality and adaptability for such problems, which is reflected in the following aspects: First, the proposed method has applicability to the supply–demand matching problem on online platforms and other multi-attribute decision-making problems. Second, the preference information chosen in this paper has a variety of forms, which can describe multi-attribute matching decision-making problems in multiple forms. Third, the proposed method considers the intermediary benefit, making the research on supply–demand matching decision-making more realistic. Finally, this paper introduces risk aversion coefficients and weight balance coefficients, which allows for adjustment according to actual situations, further improving the applicability of the method proposed in this paper.

7 Conclusion

A decision method for one-to-many matching on a logistics O2O platform considering multiple fuzzy preference information and the intermediary benefit is proposed in this paper. The method utilizes cloud model to process linguistic term set information and subsequently normalizes the multiple fuzzy preference information provided by two-sided agents of the logistics O2O platform. The method that combines the BWM with maximum deviation is used to calculate attributes weights. Based on prospect values and attribute weights, the expected matching matrix is calculated. Furthermore, the satisfaction of two-sided agents is calculated based on regret theory. Considering the objectives of agent satisfactions and the intermediary benefit, and the constraints of one-to-many matching, a multiobjective programming model is constructed. A max–min optimal algorithm considering fairness is developed to solve the model to obtain the optimal TSM scheme. The proposed model for solving multi-attribute O2O supply–demand matching problems is equally applicable to other two-sided matching decision problems. For different matching problems, corresponding matching models can be established according to the decision steps proposed in this paper, and then be solved to obtain the optimal matching results.

Main innovations of this paper are as follows: (1) The proposed decision method considers the different psychological behavioral factors of two-sided agents in the decision stage of the logistics O2O platform. (2) A combination of the BWM and maximum deviation method is proposed for calculating the weights of attributes. (3) A calculation method for the intermediary benefit considering the risk preference is provided. (4) A type of one-to-many TSM model considering the intermediary benefit objectives is constructed. (5) A max–min optimal algorithm considering fairness is developed.

However, the TSM decision method proposed in this paper still has limitations. First, in addition to the truck-load and less-than-truck-load (LTL) matching methods, logistics O2O platforms are receiving increased attention in many-to-many vehicle-cargo matching, which may not be adequately addressed by the vehicle-cargo matching method proposed in this paper. Second, when carriers and shippers engage in matching transactions through logistics O2O platforms, they typically rely on information pertaining to the attributes of vehicles and cargo. However, the model developed in this paper does not consider the influence of vehicle and cargo attributes on the matching results. Third, to validate the effectiveness and feasibility of the proposed methodology, this paper conducted calculations were conducted using a small-scale data sample. However, when handling with larger datasets, the algorithm employed in this paper to solve the model may not be capable of obtaining the optimal solution quickly and accurately.

Future research directions are summarized as follows: (1) To address TSM problems on the O2O platform in various scenarios, it would be meaningful to conduct research on many-to-many TSM problems under bounded rationality. (2) In the newly developed two-sided matching methods, a comprehensive consideration of vehicle and cargo attribute information, as well as preference evaluation information, is needed. (3) For the constructed multiobjective optimization model, new intelligent optimization algorithms need to be developed to meet the demand for vehicle-cargo matching in situations with large data volumes.

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Declarations

Conflicts of interest The authors declare no competing interests.

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